

Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.18a)

Metric

$$ds^2 = \left(-\frac{\left(\frac{\sin(\frac{1}{2}t)}{\cos(\frac{1}{2}t)} \right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2}{\lambda^2} \right) dt \otimes dt + \left(\frac{1}{\lambda^2 \left(\frac{\sin(\frac{1}{2}t)}{\cos(\frac{1}{2}t)} \right)^{2\sqrt{-M\lambda^4+1}}} \right) dx \otimes dx \\ + \left(\frac{\left(\frac{\sin(\frac{1}{2}t)}{\cos(\frac{1}{2}t)} \right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2}{\lambda^2} \right) dy \otimes dy + \left(\frac{\left(\frac{\sin(\frac{1}{2}t)}{\cos(\frac{1}{2}t)} \right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2 \sin(y)^2}{\lambda^2} \right) dz \otimes dz$$

Coordinates

$$(x^\mu) = (t, x, y, z)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{2} \tan(\frac{1}{2}t)^{(\sqrt{-M\lambda^4+1})} \sin(t)}{2\lambda} \right) dt + \left(\frac{\sqrt{2} \tan(\frac{1}{2}t)^{-\sqrt{-M\lambda^4+1}}}{2\lambda} \right) dx$$

$$n = \left(\frac{\sqrt{2} \tan(\frac{1}{2}t)^{(\sqrt{-M\lambda^4+1})} \sin(t)}{2\lambda} \right) dt + \left(-\frac{\sqrt{2} \tan(\frac{1}{2}t)^{-\sqrt{-M\lambda^4+1}}}{2\lambda} \right) dx$$

$$m = \left(\frac{\sqrt{2} \tan(\frac{1}{2}t)^{(\sqrt{-M\lambda^4+1})} \sin(t)}{2\lambda} \right) dy + \left(\frac{i \sqrt{2} \tan(\frac{1}{2}t)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)}{2\lambda} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2} \tan \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)}{2\lambda} \right) dy + \left(-\frac{i \sqrt{2} \tan \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)}{2\lambda} \right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \right) dt + \left(\frac{\sqrt{2}\lambda \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}} \right) dx$$

$$n = \left(-\frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \right) dt + \left(-\frac{\sqrt{2}\lambda \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}} \right) dx$$

$$m = \left(\frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \right) dy + \left(\frac{i \sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)} \right) dz$$

$$\bar{m} = \left(\frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \right) dy + \left(-\frac{i \sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)} \right) dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \partial_t + \frac{\sqrt{2}\lambda \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}} \partial_x$$

$$\Delta = -\frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \partial_t - \frac{\sqrt{2}\lambda \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}} \partial_x$$

$$\delta = \frac{\sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \partial_y + \frac{i \sqrt{2}\lambda \cos \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin \left(\frac{1}{2} t \right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)} \partial_z$$

$$\bar{\delta} = \frac{\sqrt{2}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin(t)} \partial_y - \frac{i \sqrt{2}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})}}{2 \sin\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)} \partial_z$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin\left(\frac{1}{2}t\right)^{-\sqrt{-M\lambda^4+1}} \cos(t)}{2 \sin(t)^2} + \frac{\sqrt{2}\sqrt{-M\lambda^4+1} \lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin\left(\frac{1}{2}t\right)^{-\sqrt{-M\lambda^4+1}}}{4 \cos\left(\frac{1}{2}t\right) \sin\left(\frac{1}{2}t\right) \sin(t)}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin\left(\frac{1}{2}t\right)^{-\sqrt{-M\lambda^4+1}} \cos(t)}{2 \sin(t)^2} - \frac{\sqrt{2}\sqrt{-M\lambda^4+1} \lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin\left(\frac{1}{2}t\right)^{-\sqrt{-M\lambda^4+1}}}{4 \cos\left(\frac{1}{2}t\right) \sin\left(\frac{1}{2}t\right) \sin(t)}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \cos(y)}{4 \sin\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)}$$

$$\beta = \frac{\sqrt{2}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \cos(y)}{4 \sin\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin(t) \sin(y)}$$

$$\tau = 0$$

$$\epsilon = \frac{\sqrt{2}\sqrt{-M\lambda^4+1}\lambda \cos\left(\frac{1}{2}t\right)^{(\sqrt{-M\lambda^4+1})} \sin\left(\frac{1}{2}t\right)^{-\sqrt{-M\lambda^4+1}}}{8 \cos\left(\frac{1}{2}t\right) \sin\left(\frac{1}{2}t\right) \sin(t)}$$

Ricci Tensor Components

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

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$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\begin{aligned} \Phi_{22} = & -\frac{M\lambda^6 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{8\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right) \cos(t) \sin\left(\frac{1}{2}t\right)}{4\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^3} \\ & - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^2}{4\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} + \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^4}{2\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^4} + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{8\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} \\ & - \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^2}{2\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^4} + \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{8\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} \end{aligned}$$

$$\begin{aligned} \Lambda = & \frac{M\lambda^6 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{48\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right) \cos(t) \sin\left(\frac{1}{2}t\right)}{24\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^3} \\ & + \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^2}{24\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} - \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^4}{12\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^4} - \frac{\sqrt{-M\lambda^4+1}\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{48\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} \\ & + \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^2}{12\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^4} - \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{48\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} \end{aligned}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & -\frac{M\lambda^6 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{6\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} + \frac{\sqrt{-M\lambda^4+1} \lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right) \cos(t) \sin\left(\frac{1}{2}t\right)}{3\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^3} \\
& + \frac{\sqrt{-M\lambda^4+1} \lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^2}{6\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} - \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^4}{3\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^4} - \frac{\sqrt{-M\lambda^4+1} \lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{12\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2} \\
& + \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \cos\left(\frac{1}{2}t\right)^2}{3\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^4} + \frac{\lambda^2 \cos\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}}}{6\left(\cos\left(\frac{1}{2}t\right)^4 - \cos\left(\frac{1}{2}t\right)^2\right) \sin\left(\frac{1}{2}t\right)^{2\sqrt{-M\lambda^4+1}} \sin(t)^2}
\end{aligned}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"